

Vol 16 Chapter 7 — Bergman Kernel as Matrix-Coefficient Sum: v0.5 WALK-BACK of the Szegő unification (Cal #259 + Lyra F43)

2026-06-06 Saturday ~12:45 EDT (date-verified)

Vol 16 Chapter 7 v0.5 — WALK-BACK of the Szegő unification

0. Why v0.5 exists (Cal #259 + Lyra F43)

v0.4 Section 4 asserted, and v0.4 Section 1 leaned on, an operator-unification: that the muon edge (81/8) and the Λ vacuum factor-2 “use the same Szegő factorization / projection P” — the Cal #254 “share an operator, not an integer” win. Cal #259 showed this rests on a conflation (caught in Lyra’s F38, extended by Lyra to her own Composite v0.5 in F43): “(1–P)” was used with two incompatible meanings —

- (a) orthogonal complement of the holomorphic space (additive; what ρ contains),
- (b) Shilov boundary-value realization of that space (what the Hardy isometry is about).

The Hardy isometry gives $E[P] = E[P]$ (a tautology about P); it does not give $\rho = E[(1-P)]/E[P]$. So the Λ factor-2 was never derived, and the muon edge (which needs meaning (a)) and the vacuum (which used meaning (b)) are two different decompositions, not one shared operator. v0.5 corrects v0.4 accordingly.

1. WITHDRAWN from v0.4

- Section 4 unification (“same Szegő factorization / projection P; strong-form Schur-generator; Cal #254 confirmed at the constant level”) — WITHDRAWN. The operator-sharing of muon-edge with vacuum is a CONJECTURE, not a result.
- Section 1 “81/8 FORCED” — DOWNGRADED. v0.4 derived 81/8 via “(1–P) = Shilov boundary defect,” i.e. meaning (b); but the muon edge needs meaning (a), the orthogonal complement. So the derivation used the wrong (1–P) realization. 81/8 is a standalone candidate with its own falsifier (boundary matrix element = 81/8 or not), pending re-derivation under a consistently-chosen (1–P).

2. What SURVIVES (verified)

- Shilov boundary = $S^4 \times S^1 / Z_2$ is a PRODUCT $\Rightarrow$ kernel factorizes — rigorous geometry, independent of which (1–P) meaning. The factorization structure $\kappa = \kappa_{S^4} \cdot \kappa_{S^1}$ stands.
- $81/8 = N_c^4 \cdot 2^{\{-N_c\}}$ keeps its standalone falsifier (loses only operator-sharing).
- $[H_B, P] = 0$ (additive split real, Lyra F37); the Hardy isometry is correct theory (just pointed at the wrong pair for ρ).
- F39 CKM Direction-B color² fork — independent, survives.
- The curvature reframe (Lyra F41 = Elie Toys 4005/4007): $R(k) = -C(k,2)/n_C = C(k,2)/\kappa_{\text{Bergman}}$ (since $\kappa_{\text{Bergman}} = -n_C$). So the sum-of-roots target $\Sigma \text{ roots}(a_{k(n)}) = C(k,2)/n_C = -C(k,2)/\kappa_{\text{Bergman}}$ is a curvature statement — the genuine surviving mechanism lead, tying my heat-trace closed form to the Bergman curvature. This is now the strongest standing thread from the Ch 7 work.

3. The sharpened open core

One decision, cleaner than v0.4's apparent closure: what is $(1-P)$? Choose the complement OR the boundary-realization, apply it CONSISTENTLY to both the vacuum ρ and the muon edge, and re-derive each under it. Then the unification either re-closes honestly or stays two separate decompositions. Multi-week (Cal #189).

4. Discipline note

Per Cal #27: the discipline fires hardest at peak convergence — “81/8 substrate-FORCED, sub-PCAP closed” (my ~11:55 post) was peak convergence, the wrong-result profile. Cal #259 caught it before it propagated as a false unification beyond this chapter; v0.5 is that correction. Numbered correction (Toy 4008) of Toy 4006 + Ch 7 v0.4, extending Cal's catch to my own work (parallel to Lyra's F43 self-extension).

— Elie, Saturday 2026-06-06 ~12:45 EDT (date-verified). Toy 4008 SCORE 4/4.